

Polytomous IRT Models

The IRT models for representing the relation between a continuous latent trait variable and a dichotomous item response presented in chapter 4 are limited in that many response formats commonly employed by psychologists cannot be scored as true versus false or right versus wrong. Numerous measurement instruments, especially in the attitude and personality assessment domains, include items with multiple ordered-response categories. Researchers employ these formats for a variety of reasons but mainly because they are more informative and reliable than dichotomously scored items. For these multiple-category types of item-response data, polytomous IRT models are needed to represent the nonlinear relation between examinee trait level and the probability of responding in a particular category.

In this chapter we present an overview of IRT models for describing polytomous item response data. Many polytomous IRT models are available, and new models are proposed yearly. We do not provide a comprehensive summary of extant models. Rather, we provide basic descriptions of the psychometric properties of six fairly well-known polytomous models. Readers interested in a wider variety of models and more technical details pertaining to item parameter estimation and fit assessment are directed to van der Linden and Hambleton (1996).

We begin by describing the homogeneous case of the graded response model (GRM; Samejima, 1969) that is an extension of the 2PL model discussed in the previous chapter. We then describe Muraki's (1992) modification of the graded response model for use with questionnaires that have a common rating scale format (e.g., all item responses scored on a

five-point scale). These first two models are considered as “indirect” models because a two-step process is needed to determine the conditional probability of an examinee responding in a particular category. The remaining models to be described are considered “direct” IRT models because only a single equation is needed to describe the relationship between examinee trait level and the probability of responding in a particular category. Specifically, we cover two polytomous models that are extensions of the Rasch 1PL: the partial credit model (PCM; Masters, 1982) and the rating scale model (RSM; Andrich, 1978a; 1978b). We also describe Muraki’s (1990) modification of the PCM to include an item slope parameter. Finally, we introduce a general model called the nominal response model (NRM; Bock, 1972) that is appropriate when item responses cannot necessarily be classified into ordered categories. We conclude by briefly summarizing several alternative models and with discussion of how researchers may select between competing models.

To illustrate model application, we use the responses of 350 undergraduates to the 12 items on the Neuroticism scale of the Neuroticism Extraversion Openness Five-Factor Inventory (NEO-FFI) (Costa & McCrae, 1992). In Table 5.1 for each item is displayed a brief description of the item content, the observed response frequencies within each category, and the item mean score when scored on a 0 to 4 scale. The item means do not demonstrate a wide spread, however, within items the response frequencies across the categories are quite variable. In Table 5.2 is displayed a listing of the raw score means by response category and item-test

TABLE 5.1
Content of the 12 NEO-FFI Neuroticism Items,
Response Frequencies, and Item Means ($N = 350$)

<i>Content</i>	<i>Category</i>					<i>M</i>
	<i>0</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	
1 Is not a worrier (R) ^a	29	55	50	137	79	2.52
2 Feels inferior to others	35	120	100	72	23	1.70
3 Feels like going to pieces under stress	25	72	43	141	69	2.45
4 Rarely feels lonely, blue (R)	20	90	68	125	47	2.25
5 Feels tense and jittery	17	111	97	101	24	2.01
6 Sometimes feels worthless	72	89	52	94	43	1.85
7 Rarely feels fearful or anxious (R)	11	95	79	130	35	2.24
8 Gets angry with way treated by others	22	98	92	109	29	2.07
9 Feels discouraged, like giving up	27	128	66	95	34	1.96
10 Is seldom sad, depressed (R)	22	105	79	107	37	2.09
11 Often feels helpless, needy	56	129	68	73	24	1.66
12 Experiences shame	30	88	55	118	59	2.25

Note. 0 = strongly disagree; 1 = disagree; 2 = neutral; 3 = agree; 4 = strongly agree.

^aAll items have already been reverse coded.

TABLE 5.2
The Mean Raw Total Scores as a Function of Response Category
for the 12 NEO-FFI Neuroticism Items ($N = 350$)

Item	Mean Raw Scores					
	1	2	3	4	5	6
0	23.2	15.8	14.8	17.7	14.0	17.0
1	19.5	25.7	19.1	18.4	20.4	21.6
2	21.2	26.0	21.9	23.8	25.7	25.9
3	25.2	31.0	27.4	28.8	29.3	29.8
4	32.0	34.7	32.3	33.1	34.6	34.6
R_{ps}	.43	.61	.64	.64	.59	.70

Item	Mean Raw Scores					
	7	8	9	10	11	12
0	16.4	18.0	14.3	18.5	16.2	16.9
1	19.8	21.4	20.3	19.7	22.5	20.5
2	22.9	25.1	26.0	24.3	27.1	23.9
3	28.9	28.7	30.1	29.5	31.1	27.7
4	32.7	29.4	36.0	33.1	35.7	32.1
R_{ps}	.56	.42	.72	.58	.67	.56

Note. Item responses are scored on a 0 to 4 scale.

polyserial correlations for each item. These values demonstrate substantial differences among the NEO-FFI items in terms of their relationship with the total test score and ultimately their capacity to discriminate among examinees. In all analyses to follow, we assume that the NEO-FFI item responses form a unidimensional scale and are locally independent. In practice, these assumptions would need to be supported by empirical evidence.

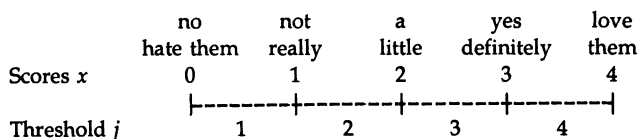
THE GRADED-RESPONSE MODEL

The graded-response model (GRM; Samejima, 1969; 1996) is appropriate to use when item responses can be characterized as ordered categorical responses such as exist in Likert rating scales. The GRM is a generalization of the 2PL model described in chapter 4 and falls under the “difference model” rubric (Thissen & Steinberg, 1986). Difference models can be considered as “indirect” IRT models because computing the conditional probability for an examinee responding in a particular category requires a two-step process. To fit the GRM within a measure the items need not

have the same number of response categories; no complications arise in item parameter estimation or the subsequent parameter interpretation as a result of a measure having items with different response formats. This is not true for the rating scale model to be described in a later section.

In the GRM, each scale item (i) is described by one item slope parameter (α_i) and $j = 1 \dots m_i$ between category "threshold" parameters (β_{ij}). We denote $m_i + 1 = K_i$ to be equal to the number of item response categories within an item. There are two stages to computing the category response probabilities in the GRM. To understand these stages, consider a test item with $K = 5$ response options where examinees receive item scores of $x = 0 \dots 4$. With five response options, there are $m_i = 4$ thresholds ($j = 1 \dots 4$) between the response options as shown in the drawing below. One goal of fitting the GRM is to determine the location of these thresholds on the latent trait continuum.

Question: *I enjoy attending large noisy fraternity parties?*



The first step in estimating response probabilities in the GRM involves the computation of m_i curves for each item of the form given in Eq. 5.1, which is simply the 2PL function described in chapter 4. Each curve shown in Eq. 5.1 represents the probability of an examinee's raw item response (x) falling in or above a given category threshold ($j = 1 \dots m_i$) conditional on trait level (θ) (for convenience we are dropping the subscript s after θ).

$$P_{ix}^*(\theta) = \frac{\exp[\alpha_i(\theta - \beta_{ij})]}{1 + \exp[\alpha_i(\theta - \beta_{ij})]} \quad (5.1)$$

where

$$x = j = 1, \dots, m_i.$$

We refer to the $P_{ix}^*(\theta)$ curves as "operating characteristic curves." In the GRM, one operating characteristic curve must be estimated for each between category threshold, and hence for a graded response item with five categories, four β_{ij} parameters are estimated and one common item slope (α_i) parameter. The β_{ij} parameters have a simple interpretation; namely, their value represents the trait level necessary to respond above threshold

j with .50 probability. Essentially, what occurs in the GRM is that the item is treated as a series of $m_i = K - 1$ dichotomies (i.e., 0 vs. 1, 2, 3, 4; 0, 1 vs. 2, 3, 4; 0, 1, 2 vs. 3, 4; 0, 1, 2, 3, vs. 4) and 2PL models are estimated for each dichotomy with the constraint that the slopes of the operating characteristic curves are equal within an item. Once these 2PL operating characteristic curves are estimated (i.e., the $P_{ix}^*(\theta)$ are estimated), computing the actual category response probabilities for $x = 0 \dots 4$ is done by subtraction as shown in Eq. 5.2:

$$P_{ix}(\theta) = P_{ix}^*(\theta) - P_{i(x+1)}^*(\theta) \quad (5.2)$$

By definition, the probability of responding in or above the lowest category is $P_{i0}^*(\theta) = 1.0$, and the probability of responding above the highest category is $P_{i5}^*(\theta) = 0.0$. Keeping with the present example, the probability of responding in each of the five categories is as follows:

$$\begin{aligned} P_{i0}(\theta) &= 1.0 - P_{i1}^*(\theta) \\ P_{i1}(\theta) &= P_{i1}^*(\theta) - P_{i2}^*(\theta) \\ P_{i2}(\theta) &= P_{i2}^*(\theta) - P_{i3}^*(\theta) \\ P_{i3}(\theta) &= P_{i3}^*(\theta) - P_{i4}^*(\theta) \\ P_{i4}(\theta) &= P_{i4}^*(\theta) - 0 \end{aligned}$$

We call these curves *category response curves* (CRCs), and they represent the probability of an examinee responding in a particular category conditional on trait level. For illustrative purposes, in Fig. 5.1 is shown the four operating characteristic curves, $P_{ix}^*(\theta)$, for a graded response item with five categories and $\alpha_i = 1.5$, $\beta_{i1} = -1.5$, $\beta_{i2} = -0.5$, $\beta_{i3} = 0.5$, and $\beta_{i4} = 1.5$. From this figure, it is evident that the between category threshold parameters represent the point along the latent trait scale at which examinees have a .50 probability of responding above a category threshold. Equivalently, the β_{ij} parameters represent the point on the latent trait scale where examinees have a .50 probability of responding in or above category $j = x$. In Fig. 5.2 are shown the category response curves for this item. These curves represent the probability of responding in each category ($x = 0 \dots 4$) conditional on examinee trait level. Observe that for any fixed value of θ , the sum of the response probabilities is equal to 1.0.

The item parameters in the GRM dictate the shape and location of the category response curves and operating characteristic curves. In general, the higher the slope parameters (α_i), the steeper the operating characteristic curves (Fig. 5.1) and the more narrow and peaked the category response curves (Fig. 5.2) indicating that the response categories differentiate among trait levels fairly well. The between category threshold parameters (β_{ij}) determine the location of the operating characteristic

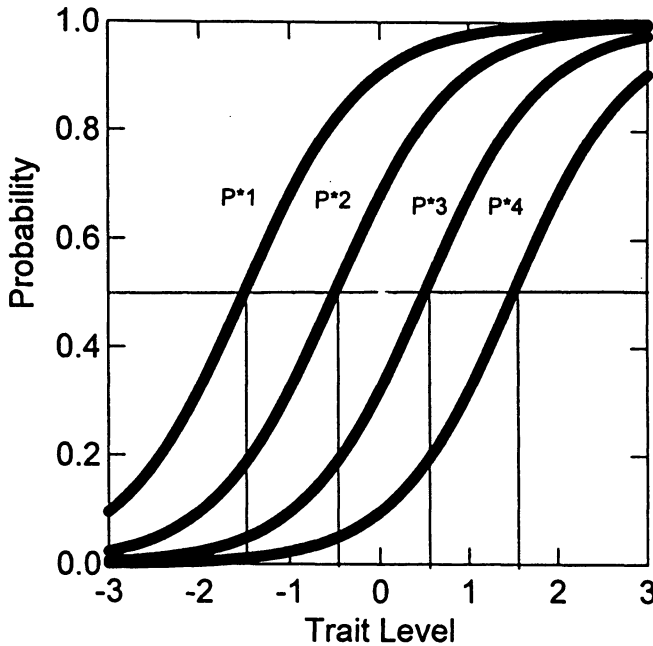


FIG. 5.1. Operating characteristic curves for five-category example item under the graded response model.

curves (Fig. 5.1) and where each of the category response curves (Fig. 5.2) for the middle response options peaks. Specifically, the category response curves peak in the middle of two adjacent threshold parameters. Other examples of item parameter estimates for the GRM based on real data can be found in Koch (1983), Reise, Widaman, & Pugh (1993), and Steinberg & Thissen (1995).

In Table 5.3 are shown the estimated item parameters for the 12 NEO-FFI neuroticism items using the program MULTILOG (Thissen, 1991). The between category threshold parameters appear to be fairly well spread out over the trait range. For some items the first and last category threshold parameters are not estimated well (e.g., Items 1 and 8). This occurs because few examinees select the extreme options for this item as shown in Table 5.1 and these items are not highly related to the latent trait. Notice that within each item, the between category threshold parameters are ordered. This must occur in the GRM. As demonstrated shortly, however, the property of ordered threshold parameters is not a requirement in the partial credit or generalized partial credit models.

Item 9 ("feels discouraged, like giving up") and Item 6 ("sometimes feels worthless") have the largest slope parameters while Item 8 ("gets angry with way others treat them") and Item 1 ("is not a worrier") have the

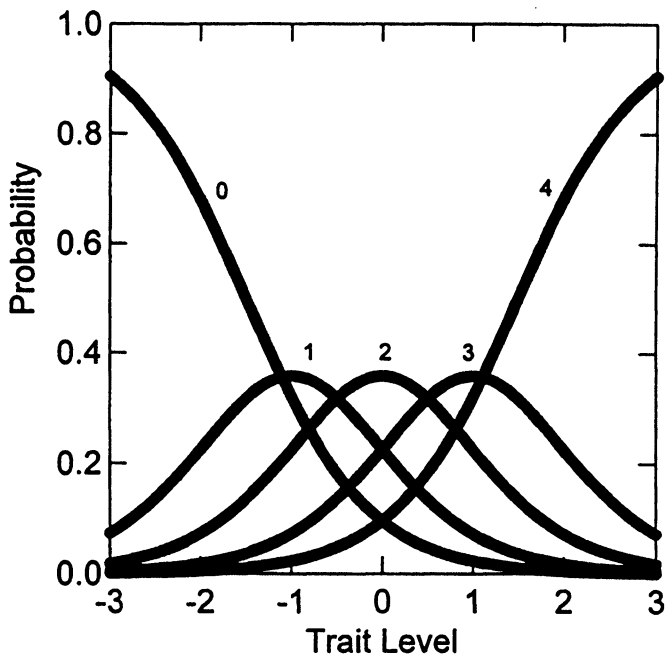


FIG. 5.2. Category response curves for five-category example item under the graded response model.

TABLE 5.3
Estimated Item Parameters for the Graded Response Model

	α (SE)	β_1 (SE)	β_2 (SE)	β_3 (SE)	β_4 (SE)
1	0.70 (0.13)	-3.80 (0.84)	-1.93 (0.43)	-0.87 (0.28)	1.88 (0.39)
2	1.42 (0.15)	-2.07 (0.25)	-0.22 (0.12)	0.93 (0.15)	2.42 (0.26)
3	1.43 (0.15)	-2.37 (0.28)	-0.93 (0.14)	-0.39 (0.13)	1.34 (0.17)
4	1.31 (0.15)	-2.72 (0.36)	-0.81 (0.15)	0.04 (0.13)	1.85 (0.24)
5	1.14 (0.14)	-3.14 (0.42)	-0.60 (0.16)	0.64 (0.15)	2.72 (0.39)
6	1.84 (0.19)	-1.15 (0.14)	-0.15 (0.10)	0.37 (0.10)	1.60 (0.18)
7	1.06 (0.13)	-3.75 (0.57)	-0.99 (0.20)	0.11 (0.16)	2.47 (0.37)
8	0.65 (0.12)	-4.43 (0.90)	-1.08 (0.31)	0.75 (0.28)	3.96 (0.79)
9	2.09 (0.20)	-1.93 (0.18)	-0.20 (0.09)	0.42 (0.09)	1.70 (0.17)
10	1.18 (0.14)	-2.81 (0.39)	-0.64 (0.16)	0.37 (0.15)	2.24 (0.32)
11	1.69 (0.18)	-1.46 (0.17)	0.08 (0.10)	0.81 (0.12)	2.13 (0.23)
12	1.15 (0.14)	-2.52 (0.35)	-0.76 (0.16)	-0.04 (0.14)	1.71 (0.24)

Note. 0 = strongly disagree; 1 = disagree; 2 = neutral; 3 = agree; 4 = strongly agree.

smallest slope parameters. As will be explained in more detail shortly, in polytomous IRT models these values should not be interpreted directly as item discriminations. To directly assess the amount of discrimination these items provide a researcher needs to compute item information curves (IICs), as detailed in chapter 7.

As to be described in chapter 9, the quality of a particular model application may be examined in many ways. In this regard, a useful feature of MULTILOG is that for each item the program provides both the observed proportion of responses in each category and model predicted values (i.e., predicted by the item parameters and the estimated latent trait distribution). The difference in these values indicate how well the model predicts the actual item responses. In the present case, the estimated model parameters do an outstanding job in predicting the observed response proportions—no residual is greater than .01, and most are zero. However, these residual values would be expected to increase when the model is applied in a new sample of examinees not included in the calibration. Also, one fit index is seldom satisfactory in judging the complex issue of model-fit in IRT, and in practice, several approaches should be used.

THE MODIFIED GRADED RESPONSE MODEL

Muraki (1990) developed a modification of the GRM that facilitates the use of this model in the analysis of questionnaires with rating-scale type response formats (e.g., an attitude questionnaire where all the items have the same number of response categories). Herein we refer to this model as the “modified” graded response model (M-GRM). Note that Muraki and others refer to this model as a “rating scale model.” We will not use this nomenclature because we believe that it might confuse readers into thinking that the M-GRM is equivalent to the Andrich rating scale model to be described shortly. In fact, the M-GRM is a restricted case of Samejima’s (1969) GRM described in the previous section. Like the GRM, the M-GRM allows the item-slope parameters to vary across items and is indirect (i.e., computing response probabilities requires a two-step process). However, in the M-GRM the between category threshold parameters (β_{ij}) of the GRM are partitioned into two terms, namely, a location parameter (b_i) for each item, and a set of category threshold parameters (c_j) for the entire scale. That is, $\beta_{ij} = b_i - c_j$.

The operating characteristic curves for the M-GRM model can be written as:

$$P_{ix}^*(\theta) = \frac{\exp[\alpha_i(\theta - (b_i - c_j))]}{1 + \exp[\alpha_i(\theta - (b_i - c_j))]} \quad (5.3)$$

or equivalently,

$$P_{ix}^*(\theta) = \frac{\exp[\alpha_i(\theta - b_i + c_j)]}{1 + \exp[\alpha_i(\theta - b_i + c_j)]} \quad (5.4)$$

and therefore the probability of responding in a particular category is:

$$P_{ix}(\theta) = P_{ix}^* - P_{i(x+1)}^* \quad (5.5)$$

where, just as with the GRM,

$$P_{i(x=0)}^*(\theta) = 1.0 \quad \text{and} \quad P_{i(x=m+1)}^*(\theta) = 0.0$$

Equation 5.3 shows the probability of responding in a given category ($x = 1, \dots, m_i$) or higher and is analogous to Eq. 5.1. The slope parameter in the M-GRM is analogous to that in the GRM; it indicates how quickly the expected item scores change as a function of trait level. The difference between the GRM and M-GRM is that in the GRM one set of category threshold parameters (β_{ij}) is estimated for each scale item, whereas in the M-GRM one set of category threshold parameters (c_j) is estimated for the entire scale, and one location parameter (b_i) is estimated for each item. As can be seen from Eq. 5.3 the item location parameter serves to move the category threshold parameters within an item up and down the trait continuum. These location parameters thus indicate the "difficulty," or scale value, of a particular item. Few examinees are expected to score in the highest response categories on an item with a large positive b_i value, and conversely, few examinees are expected to score low on an item with a large negative b_i value. In sum, the M-GRM is "restricted" because the model assumes that the category boundaries are equally distant from each other across scale items, whereas in the GRM they are free to vary across items. As a consequence, the M-GRM requires fewer parameters estimates than the GRM.

The motivation for the M-GRM model was to describe responses to Likert format attitude scales (Muraki, 1990). One advantage of the M-GRM, relative to the GRM, is that it allows separation of the estimation of item location and category threshold parameters. However, note that there is an arbitrariness to the scale of the category threshold parameters (c_j). If a measurement instrument contains items with different response formats, then the GRM is much easier to implement in practice relative to the M-GRM. If the M-GRM were to be estimated for items with different numbers of response categories, then items with similar formats would have to be treated as a "block" and category and item parameters estimated within blocks. In this case, the item parameters (α_i and b_i) within a block could not be directly compared to item parameters in other blocks

because of the indeterminacy of the scale for the c_j parameters. To compare item parameters across blocks, a researcher would have to place the item parameters onto a common metric using some form of scale linking. On the other hand, if a set of scale items has a common set of response options, the M-GRM may provide advantages over the GRM in that the item locations and the category threshold parameters can be estimated separately. One advantage of this is that the item location parameters (b_i) can be used to order the items according to their difficulty (or scale value). Also, the category threshold parameters (c_j) provide an estimate of the psychological distance between the scale points independent of the item parameters (Muraki, 1990). These properties combined make the M-GRM analogous to Thurstone's method of successive categories.

The neuroticism item parameters (i.e., slope and location) under the M-GRM are shown in the first two columns of Table 5.4. These estimates were derived using PARSCALE (Muraki, 1993) and all twelve items were treated as a single block. [Note that when the analysis was conducted treating each item as its own block, parameter estimates were almost exactly the same as for the GRM estimated with MULTILOG.] Thus, any differences in the parameters given in Tables 5.3 and 5.4 are due solely to the estimation of a "common" set of category thresholds in the M-GRM. On the bottom of Table 5.4 is shown the category threshold parameters (c_j) for the entire scale. Notice that these thresholds in the M-GRM add up to zero and are ordered. The slope parameters in the M-GRM tend to be less vari-

TABLE 5.4
Estimated Item Parameters for the Modified Graded
Response Model and Item Chi-Square Fit Statistics

	<i>Slope (SE)</i>	<i>Location (SE)</i>	<i>CHI</i>	<i>df</i>	<i>p</i>
1	0.93 (0.05)	-0.96 (0.10)	20.66	12	0.055
2	1.45 (0.07)	0.28 (0.06)	14.67	13	0.328
3	1.13 (0.06)	-0.73 (0.08)	21.15	12	0.048
4	1.33 (0.07)	-0.39 (0.07)	6.90	11	0.808
5	1.58 (0.08)	-0.02 (0.06)	8.37	12	0.756
6	0.98 (0.05)	0.25 (0.09)	36.90	13	0.000
7	1.51 (0.08)	-0.35 (0.06)	11.10	10	0.349
8	1.19 (0.06)	-0.09 (0.08)	18.47	13	0.140
9	1.50 (0.08)	0.06 (0.06)	21.37	13	0.060
10	1.33 (0.07)	-0.15 (0.07)	7.97	13	0.847
11	1.30 (0.07)	0.51 (0.07)	36.64	12	0.000
12	1.03 (0.05)	-0.42 (0.09)	8.28	13	0.825

M-GRM category thresholds = 2.266 (.04) 0.431 (.02) -0.451 (.02) -2.254 (.04)

Note. Taking the location (b_i) minus the category thresholds (c_j) roughly yields the β_{ij} parameters in the graded response model. They are not exactly the same because the two models make different assumptions.

and are ordered. The slope parameters in the M-GRM tend to be less variable than the slope estimates under the GRM. Even a relatively poor functioning item such as Item 1 has a slope parameter of 0.93. To conclude this section, in the last set of columns in Table 5.4 is shown the chi-square fit statistics (see chap. 9) as output from PARSCALE. These statistics indicate that 3 of the 12 items are not well represented by the estimated M-GRM item parameters ($p < .05$).

THE PARTIAL CREDIT MODEL

The partial credit model (PCM; Masters, 1982) was originally developed for analyzing test items that require multiple steps and for which it is important to assign partial credit for completing several steps in the solution process. Thus, the PCM model lends itself naturally to describing item responses to achievement tests (e.g., math problems) where partially correct answers are possible. The PCM is also perfectly appropriate for analyzing attitude or personality scale responses where subjects rate their beliefs, or respond to statements on a multi-point scale (see Masters & Wright, 1996).

In contrast to the GRM and M-GRM just described, the partial credit model is a divide-by-total (Thissen & Steinberg, 1986) or as we term it, a "direct" IRT model. This means that the probability of responding in a particular category will be written directly as an exponential divided by the sum of exponentials. The PCM can be considered as an extension of the 1PL model described in chapter 4, and it has all the standard Rasch model features such as separability of person and item parameters (see chap. 8 for details). Assume as before that item i is scored $x = 0, \dots, m_i$ for an item with $K_i = m_i + 1$ response categories. For $x = j$ the category response curves for the PCM can be written as:

$$P_{ix}(\theta) = \frac{\exp\left[\sum_{j=0}^x (\theta - \delta_{ij})\right]}{\sum_{r=0}^{m_i} \left[\exp \sum_{j=0}^r (\theta - \delta_{ij})\right]} \quad (5.6)$$

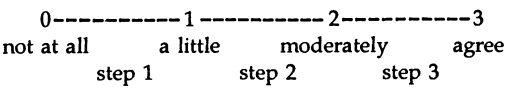
where

$$\sum_{j=0}^0 (\theta - \delta_{ij}) \equiv 0$$

The δ_{ij} ($j = 1, \dots, m_i$) term is sometimes called the item *step difficulty* associated with a category score of j ; the higher the value of a particular δ_{ij} , the more difficult a particular step is relative to other steps within an item.

A δ_{ij} term can also be directly interpreted as the point on the latent trait scale at which two consecutive category response curves intersect. For this reason, herein we refer to the δ_{ij} parameters of the PCM as *category intersections*. In the PCM all items are assumed to have equal slopes and thus this term disappears from the model. Equation 5.6 states that the probability of an examinee responding in category x on an m_i step item is a function of the difference between an examinee's trait level and a category intersection parameter. In other words, the δ_{ij} intersection parameters can be considered as step "difficulties" associated with the transition from one category to the next, and there are m_i step difficulties (intersections) for an item with $m_i + 1$ response categories.

To understand the PCM model, consider the following graphical representation of a four-category attitude-type item:



In this item, an examinee must complete three steps in order to respond in the highest category, that is, they have to decide between not at all and a little (Step 1), between a little and moderately (Step 2), and between moderately and agree (Step 3). The δ_{ij} parameters in polytomous Rasch models, such as the PCM, do not represent a point along the latent trait continuum at which an examinee has a .50 probability of responding above a category threshold, as the β_{ij} parameters do in the GRM. Rather, in the PCM the δ_{ij} parameters represent the relative difficulty of each step. Within an item some steps (category intersections) may be relatively easier or more difficult than others. Furthermore, the δ_{ij} parameters indicate where on the latent-trait continuum the category response curves intersect and hence indicate where on the latent-trait scale the response of one category becomes relatively more likely than the previous category.

To illustrate the PCM, Fig. 5.3 shows the category response curves for a three-category item (2 steps) taken from Masters (1985, p. 76). For this item, the δ_{ij} parameters are $\delta_{i1} = -0.29$ and $\delta_{i2} = 0.95$, respectively. Observe how these points represent where on the latent-trait continuum the category response curves intersect (i.e., their transitions). At these points, the completion of another step becomes relatively more likely than noncompletion given that an examinee has already completed the previous step(s). Furthermore, these category response curves can be used to compute the most likely response for examinees at various points along the trait continuum. For example, judging from Fig. 5.3 the most likely response for an examinee with a trait level of -2.0 would be to complete 0 steps (score 0), whereas the most likely response for an examinee with a trait level of 1.5 would be to complete 2 steps (Score 2). Other examples

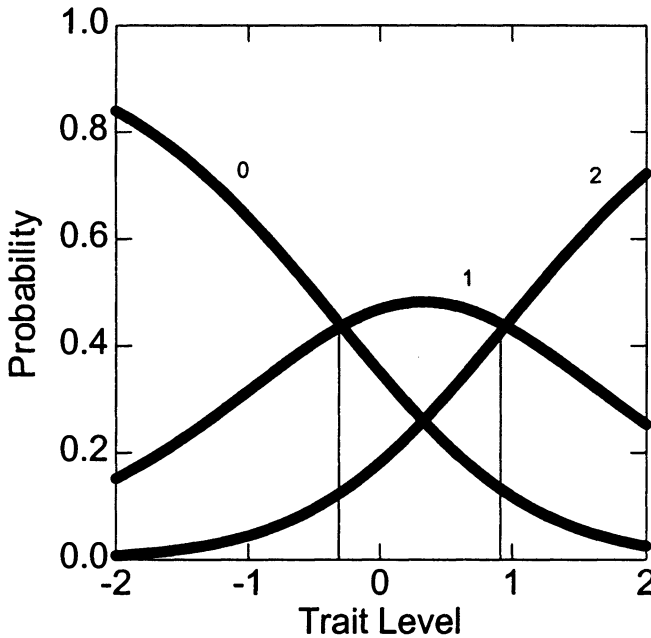


FIG. 5.3. Category response curves for the example items under the partial credit model.

and analyses of the partial credit model may be found in Masters (1984) and Wright and Masters (1982).

A second way to describe the relationship between examinee trait level and item responses is to use the PCM item parameters to graph the expected score (sometimes called a true score) on an item for all latent trait levels. This can be done by using Eq. 5.7. Such curves represent the expected raw item score for examinees at a particular trait level. If these curves are added together over items, then the resulting curve would represent the expected raw score for examinees at a particular trait level. Use of these curves will be illustrated shortly.

$$E(X) = \sum_{x=0}^{m_i} xP_x(\theta) \quad (5.7)$$

A particularly attractive feature of the PCM, as with any Rasch model, is that the raw scale score is a sufficient statistic for estimating examinee trait level; examinees with the same raw score on a set of items that fit the PCM are estimated to have equivalent positions on the latent trait. The drawback, of course, is that all items must be equally related to the underlying latent trait, a property sometimes hard to produce in practice. In the

present case with the 12 neuroticism items, we do not expect a good fit to the PCM because of the large variation in the item slope parameters found with the GRM (Table 5.3).

In Table 5.5 are shown the estimated category intersection parameters for the 12 neuroticism items. These parameters were estimated using PARSCALE (Muraki, 1993). The first observation to be drawn is that the δ_{ij} parameters are not necessarily ordered as they were for the GRM (Table 5.3). When the intersection parameters are not ordered, this phenomenon is known as a "reversal" (Dodd & Koch, 1987). To illustrate what happens when there is a reversal the CRCs for Item 3 ("feels like going to pieces under stress") is displayed in Fig. 5.4. In this item the relative order of the intersections (step difficulty) is: Step 1, Step 3, Step 2, and Step 4. In fact, what these values indicate is that going from a raw score of 0 to 1 (Step 1) or from 2 to 3 (Step 3) are relatively easy transitions for examinees to make. Notice how the PCM nicely accounts for the fact that on Item 3 category two is sel-

TABLE 5.5
Estimated Item Parameters for the Partial
Credit Model and Item-Fit Statistics

	δ_1	δ_2	δ_3	δ_4	χ^2	DF	Probability
1	-1.400 (.23)	-0.279 (.19)	-1.017 (.16)	0.923 (.14)	38.78	10	0.000
2	-1.763 (.19)	0.080 (.14)	0.622 (.15)	1.830 (.24)	11.36	11	0.413
3	-1.800 (.23)	0.167 (.19)	-1.168 (.17)	1.117 (.15)	9.36	10	0.498
4	-2.205 (.25)	-0.003 (.16)	-0.507 (.15)	1.471 (.17)	8.82	11	0.639
5	-2.519 (.26)	-0.063 (.14)	0.170 (.14)	2.055 (.23)	9.90	10	0.450
6	-0.686 (.16)	0.431 (.17)	-0.354 (.17)	1.376 (.19)	5.74	10	0.837
7	-2.890 (.32)	-0.105 (.15)	-0.383 (.14)	1.835 (.19)	13.86	10	0.179
8	-2.143 (.24)	-0.154 (.15)	0.011 (.14)	1.907 (.21)	29.31	10	0.001
9	-2.132 (.21)	0.505 (.15)	-0.139 (.16)	1.636 (.20)	7.53	11	0.755
10	-2.206 (.24)	0.065 (.15)	-0.134 (.15)	1.623 (.19)	13.35	10	0.204
11	-1.281 (.16)	0.600 (.15)	0.264 (.17)	1.823 (.24)	22.29	11	0.022
12	-1.738 (.21)	0.203 (.17)	-0.666 (.16)	1.166 (.16)	20.02	10	0.029
					190.38	124	0.000

Note. $-2 \log \text{likelihood} = 11,553.011$.

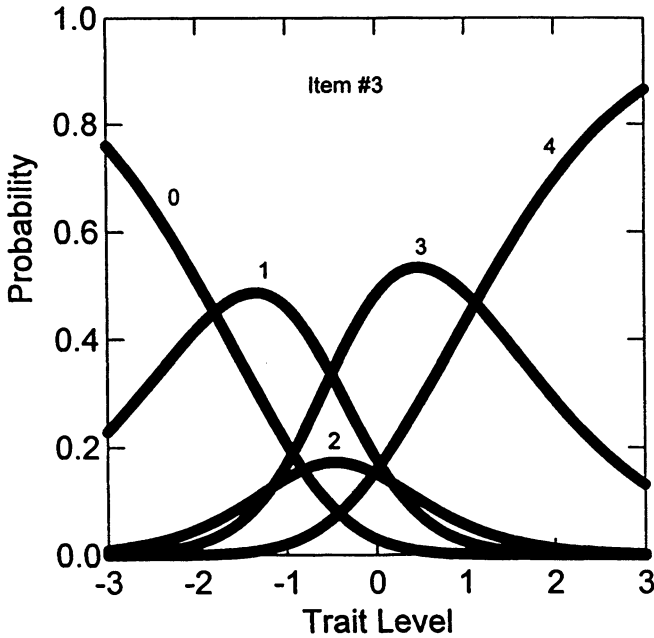


FIG. 5.4. Category response curves for Item 3 (reversal) under the partial credit model.

dom selected and responses in category three are relatively frequent (see Table 5.1).

In contrast, Fig. 5.5 shows the CRCs for Item 5 ("feels tense and jittery"). This item has ordered intersection parameters and most examinees responded in either category one, two, or three (see Table 5.1). As a rule, if the category intersection parameters are ordered within an item, then there is at least one trait level where every response option is most likely. If there is a "reversal" in intersection parameters, this guarantees that there will be at least one category that is never the most likely option conditional on trait level (Andrich, 1988).

In the last set of columns in Table 5.5 is shown the likelihood-ratio chi-square fit statistics (see chap. 9) as output from PARSCALE. These statistics indicate that 4 of the 12 items are not well represented by the estimated PCM item parameters. These chi-square statistics are additive across items and the resulting chi-square of 190.39 on 124 degrees of freedom also indicates that the overall model does not fit well. The -2 times the log-likelihood value is 11,553.011. This statistic cannot be used to assess model-fit, nevertheless it will serve a useful purpose in comparing the PCM with the generalized partial credit model described in the next section.

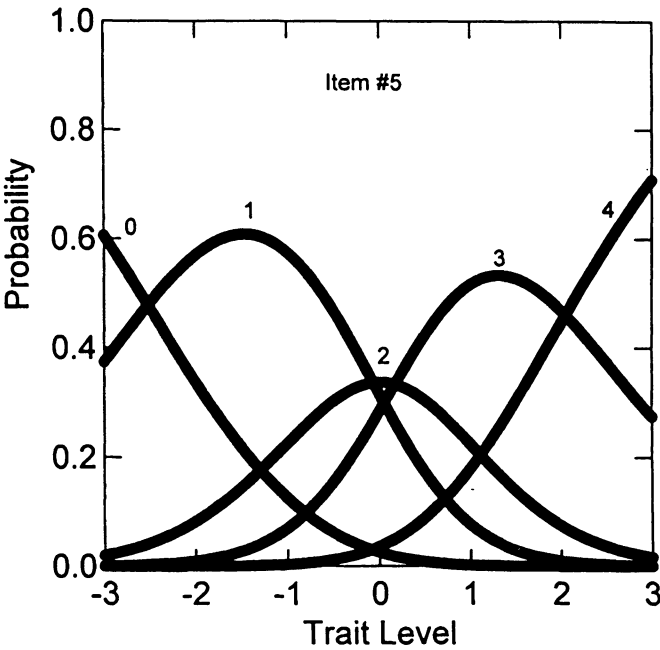


FIG. 5.5. Category response curves for Item 5 (no reversal) under the partial credit model.

Finally, as mentioned before, Eq. 5.7 shows how an examinee’s predicted item response (sometimes called an item true score) changes as a function of trait level. Because the PCM is a Rasch model, and all items have the same slope (1.0), these curves are highly similar across items. These item “true-score” curves are also additive across items, and Fig. 5.6 shows the aggregation of the 12 neuroticism true-score curves. This curve displays how the expected raw score changes as a function of trait level. Apparently, examinees who are around zero on the latent-trait variable are expected to score roughly around 24 raw score points. Inspection of such curves is one way to study the quality of a measurement instrument in terms of assessing a latent-trait variable. The curve in Fig. 5.6, which shows a nonlinear relationship between observed scores and trait levels, should be contrasted with the CTT perspective, which assumes a linear relationship.

THE GENERALIZED PARTIAL CREDIT MODEL

Muraki (1992; 1993) developed a generalization of the PCM that allows the items within a scale to differ in slope parameter. In this chapter his model will be referred to as the generalized partial credit model (G-PCM).

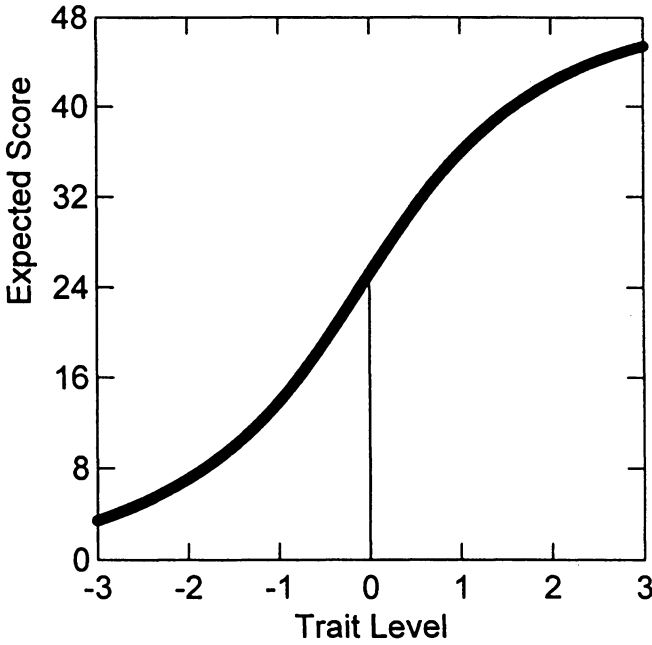


FIG. 5.6. True score curve for the neuroticism scale under the partial credit model.

Parameters for this model can be estimated using the program PARSCALE (Muraki, 1993). In Eq. 5.8 we have written the G-PCM by substituting a slope parameter into Eq. 5.6.

$$P_{ix}(\theta) = \frac{\exp \sum_{j=0}^x \alpha_i(\theta - \delta_{ij})}{\sum_{r=0}^M \left[\exp \sum_{j=0}^r \alpha_i(\theta - \delta_{ij}) \right]} \quad (5.8)$$

where

$$\sum_{j=0}^0 \alpha_i(\theta - \delta_{ij}) \equiv 0$$

The category intersection parameters (δ_{ij}) in this model are interpreted in the same way as in the PCM, that is, as the intersection point of two adjacent category response curves. They are the points on the latent-trait scale where one category response becomes relatively more likely than the preceding response, given that the examinee has completed previous steps. However, the slope (α_i) parameters are not interpreted in the same

way as they are in dichotomous IRT models. The reason is because in polytomous models the item discrimination depends on a combination of the slope parameter and the spread of the category thresholds (in GRM or M-GRM) or category intersections (in PCM and G-PCM). In the G-PCM the slope parameters "indicate the degree to which categorical responses vary among items as θ level changes" (Muraki, 1992, p. 162). As the slope parameter in Eq. 5.8 becomes smaller than 1.0 the CRCs flatten out relative to the PCM, while as the slope parameter becomes larger than 1.0 the CRCs tend to become more peaked relative to the PCM.

Note that in the previous and following PARSCALE analyses, we specified that the 12 NEO-FFI neuroticism items each formed their own "block" (see chap. 13 for details). Consequently, a separate set of category intersection parameters is estimated for each item. The G-PCM item slope and intersection estimates for the neuroticism items are shown in Table 5.6. First, notice that there is quite a large degree of variation among the

TABLE 5.6
Estimated Item Parameters for the Generalized
Partial Credit Model and Item-Fit Statistics

	α	δ_1	δ_2	δ_3	δ_4	χ^2	DF	Probability
1	0.261 (.03)	-2.937 (.87)	0.121 (.74)	-3.857 (.63)	2.328 (.54)	36.29	13	0.001
2	0.877 (.07)	-2.130 (.22)	0.070 (.16)	0.755 (.18)	2.197 (.28)	8.71	13	0.795
3	0.797 (.06)	-2.295 (.30)	0.197 (.24)	-1.462 (.22)	1.399 (.19)	9.29	14	0.813
4	0.735 (.06)	-2.923 (.34)	0.028 (.22)	-0.703 (.20)	1.919 (.23)	3.02	13	0.998
5	0.683 (.05)	-3.513 (.38)	-0.041 (.20)	0.182 (.21)	2.808 (.33)	9.29	12	0.678
6	1.073 (.09)	-0.873 (.15)	0.358 (.16)	-0.226 (.16)	1.547 (.18)	8.71	11	0.650
7	0.583 (.05)	-4.493 (.55)	-0.004 (.26)	-0.732 (.24)	2.792 (.33)	16.93	11	0.109
8	0.345 (.03)	-4.815 (.68)	0.012 (.42)	-0.362 (.41)	4.256 (.60)	11.58	14	0.640
9	1.499 (.11)	-1.997 (.10)	0.210 (.10)	0.103 (.11)	1.627 (.14)	4.93	11	0.934
10	0.631 (.05)	-3.215 (.37)	0.195 (.24)	-0.292 (.23)	2.299 (.30)	15.07	13	0.302
11	1.059 (.09)	-1.440 (.16)	0.551 (.15)	0.397 (.16)	2.039 (.23)	22.12	11	0.023
12	0.565 (.05)	-2.628 (.37)	0.534 (.30)	-1.241 (.29)	1.720 (.28)	9.65	13	0.723
						155.64	149	0.338

Note. $-2 \log$ Likelihood = 11,384.921.

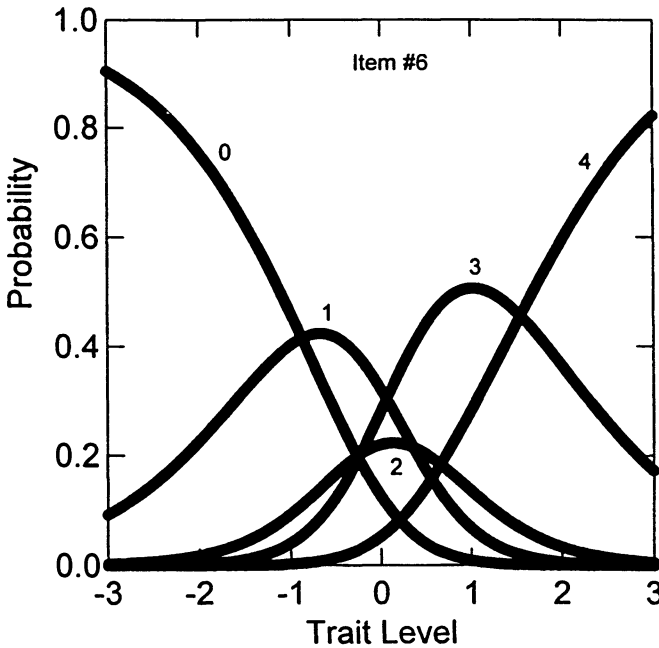


FIG. 5.7. Category response curves for Item 6 under the generalized partial credit model.

slope parameters. Generally speaking, the item ordering by slope in the G-PCM is equivalent to that observed in the GRM. To illustrate model features, we show the CRCs for three items. Figure 5.7 shows the CRCs for Item 6 ("sometimes feel worthless"). This item has a slope estimate around 1.0 and thus can be considered a baseline for comparison to other curves. Figure 5.8 displays the CRCs for Item 5 ("feels tense and jittery"), which has a relatively low slope value. Notice again how these curves do a great job of capturing the fact that most of the responses to this item occur in categories one and three. Finally, to illustrate an item with a large slope in Figure 5.9 is shown the CRCs for Item 9 ("feels discouraged, like giving up"). Observe how peaked the curves are relative to Item 5.

To conclude this section, the last set of columns in Table 5.6 shows the chi-square fit statistics output from PARSCALE. In contrast to the results with the PCM (Table 5.5), these statistics indicate a pretty good fit in that only 2 of the 12 items are not well represented by the estimated G-PCM item parameters. These chi-square statistics are additive across items and the resulting chi-square of 155.64 on 149 degrees of freedom ($p = .33$) indicates that the overall fit is adequate. The -2 times log-likelihood value is 11,553.011. Finally, the change in chi-square in going from the PCM to the

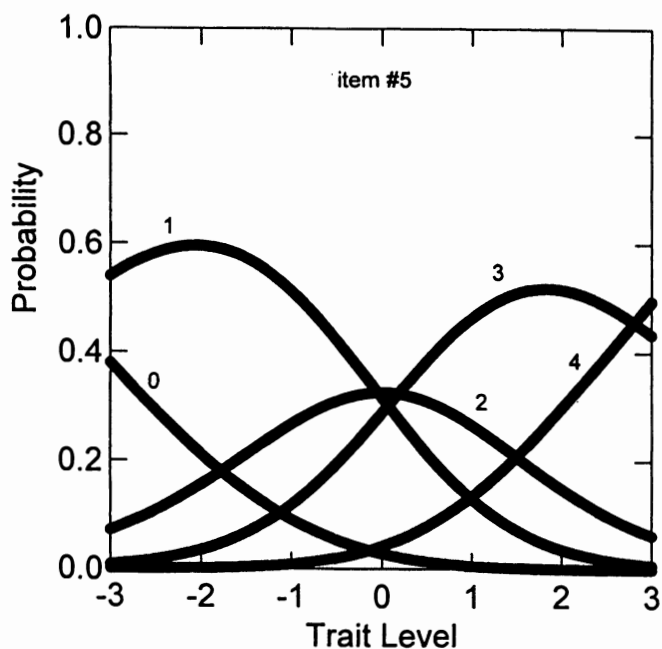


FIG. 5.8. Category response curves for Item 5 under the generalized partial credit model.

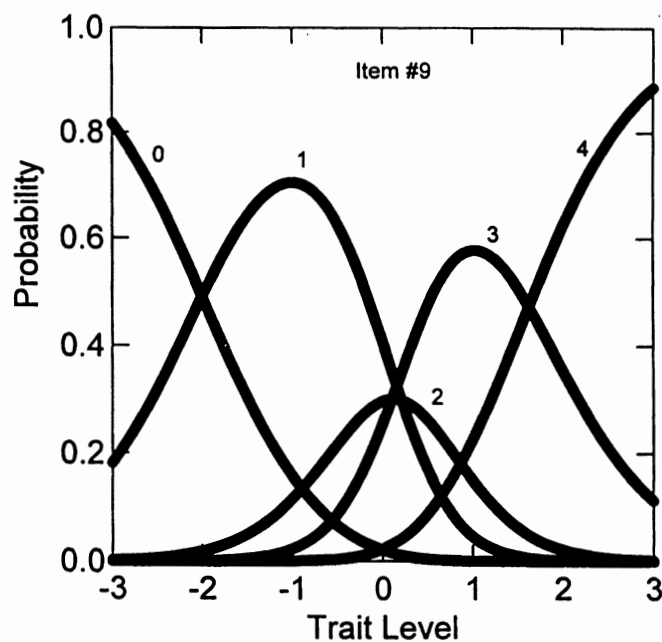


FIG. 5.9. Category response curves for Item 9 under the generalized partial credit model.

G-PCM is thus 168.08 on 12 degrees of freedom ($p < .001$), which means that the G-PCM is a significant improvement over the PCM.

RATING SCALE MODEL

Before describing the model in this section, it is important to note that the literature on the "rating scale model" can be a little confusing because there are several versions of this model that vary in complexity and are formulated in different ways by different authors (see Andersen, 1995, p. 273). Thus, what is presented in this section is just one way of conceptualizing and writing a rating scale model. With this in mind, although the rating scale model (RSM; Andrich, 1978a; 1978b) can be derived from the PCM (Masters & Wright, 1984) there is a crucial distinction. For items with the same response format, in the RSM each item is described by a single scale location parameter, λ_i , which reflects the relative easiness or difficulty of the particular item. Furthermore each of the $J = K - 1$ category thresholds, taken over all items in the measure, is described by a category intersection parameter (δ_j). In other words, response categories are assigned intersection parameters that are considered equal across items, and an item's location is described by only a single λ_i parameter (Dodd, 1990). As with the PCM described earlier, the rating scale model assumes all items to be equally discriminating, and the raw scale score is a sufficient statistic for estimating examinee trait level.

The PCM and RSM differ in that the PCM makes no assumptions about the relative difficulties of the steps (intersections) within any item. That is, the spread of the category intersection parameters can differ across different items. In contrast, with a rating scale item format where the anchors are the same across items, it is easy to assume that the relative difficulties of the steps within items should vary little. Thus in the RSM, a common set of δ_j parameters are estimated. To make this more concrete, with a set of math items that require multiple steps for completion, it makes sense that within and between items, some steps will be relatively easy while others are more difficult. However, when working with a rating scale attitude measure with common anchors, such as 1 = disagree, 2 = modestly agree, and 3 = agree completely, the relative difficulties of the steps within each item would not be expected to change much across items.

In the RSM model, the step difficulties (intersections) of the PCM (Eq. 5.6) are decomposed into two components, namely, λ_i and δ_j , where $\delta_{ij} = (\lambda_i + \delta_j)$. The λ_i is the location of the item on the latent scale, and the δ_j are the category intersection parameters. Dodd (1990) writes the RSM as Eq. 5.9.

$$P_x(\theta) = \frac{\exp\left\{\sum_{j=0}^x [\theta - (\lambda_i + \delta_j)]\right\}}{\sum_{x=0}^M \exp\left\{\sum_{j=0}^x [\theta - (\lambda_i + \delta_j)]\right\}} \quad (5.9)$$

where

$$\sum_{j=0}^0 [\theta - (\lambda_i + \delta_j)] = 0$$

Alternatively, some authors (Dodd, DeAyala, & Koch, 1995; Rost, 1988) express the RSM in a slightly different, but equivalent, form:

$$P_x(\theta) = \frac{\exp[\psi_x + x(\theta - \lambda_i)]}{\sum_{x=0}^M \exp[\psi_x + x(\theta - \lambda_i)]} \quad (5.10)$$

where

$$\begin{aligned} \psi_x &= -\sum_{j=0}^x \delta_j \\ \psi_0 &= \psi_m = 0 \end{aligned}$$

The RSM assumes that a fixed set of rating points are used for the entire item set. If the items within a scale have different formats, the rating scale model is not an appropriate choice. With a program like RUMM (Sheridan, Andrich, & Luo, 1996) however, it would be possible to estimate the parameters of groups of items with similar numbers of response categories within a single block. The measure may consists of two or three blocks of RSM items with different formats. The problem in such an analysis lies in trying to equate the item parameters across the different blocks. This is a problem for comparing items across blocks and not a problem for predicting responses, however.

In the first column of Table 5.7 are shown the estimated RSM location parameters (λ_i) for the example data set as estimated by RUMM (Sheridan, Andrich, & Luo, 1996). The location parameters in Table 5.7 indicate the "difficulty" of the item, and notice that the ordering corresponds perfectly with the item means shown in Table 5.1. That is, Item 2 has the small mean indicating that few examinees score high and Item 1 has a large mean indicating that few examinees score low on this item. Observe in Table 5.7 that Item 2 has a high location ($\lambda_2 = 0.30$), and Item 1 has a low location ($\lambda_2 = -0.44$).

On the bottom of Table 5.7 are shown the four estimated category intersection (δ_j) parameters. These values are constant across all 12 items and sum to 0. This latter feature is a scale identification constraint imposed by the program. To illustrate the effect of the location parameter, in Figs. 5.10 and 5.11 is shown the CRCs for Items 1 and 2, respectively. Notice how the

TABLE 5.7
Estimated Item Parameters for the Rating Scale Model and
Item-Fit Chi-Square Statistics Output from RUMM

	<i>Location (SE)</i>	<i>Item Fit</i>	<i>CHI</i>	<i>DF</i>	<i>Probability</i>
1	-0.44 (0.05)	7.492	9.199	1	0.031
2	0.30 (0.05)	-1.943	2.492	1	0.648
3	-0.39 (0.05)	1.008	4.470	1	0.329
4	-0.16 (0.05)	-0.934	0.960	1	0.913
5	0.08 (0.05)	-2.356	5.728	1	0.200
6	0.27 (0.05)	1.631	28.309	1	0.000
7	-0.14 (0.05)	-0.695	6.234	1	0.160
8	0.21 (0.05)	3.838	20.313	1	0.000
9	0.15 (0.05)	-3.502	7.524	1	0.087
10	0.00 (0.05)	0.010	1.318	1	0.855
11	0.47 (0.05)	-1.107	11.022	1	0.000
12	-0.17 (0.05)	3.800	1.392	1	0.841

Note. RSM intersections = -1.600, 0.224, -0.184, and 1.560.

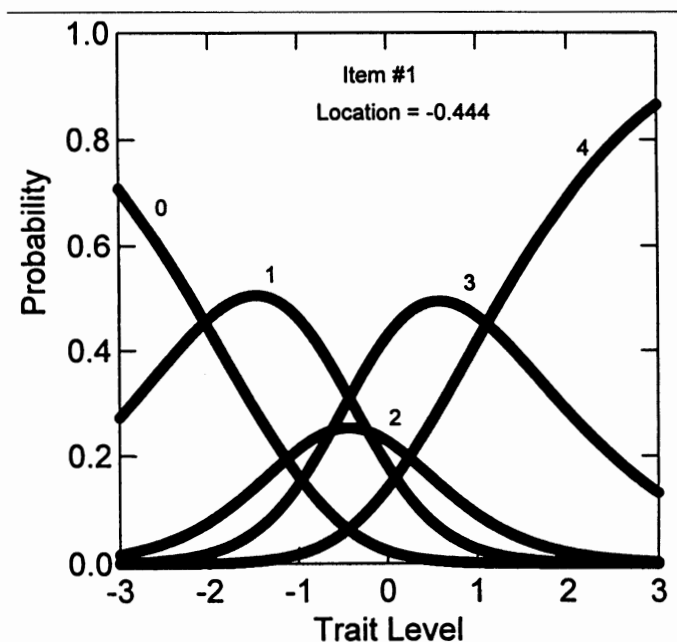


FIG. 5.10. Category response curves for Item 1 under the rating scale model.

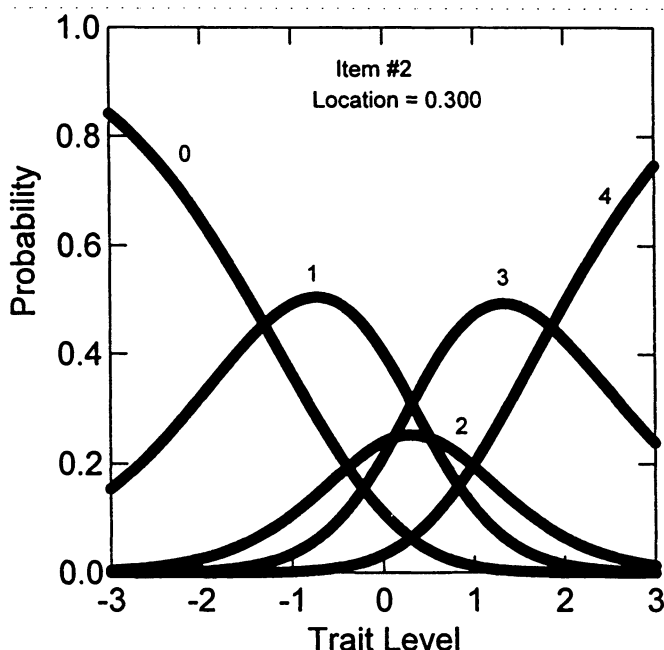


FIG. 5.11. Category response curves for Item 2 under the rating scale model.

general shape is the same but the curves for Item 2 are shifted to the right relative to Item 1. The location parameter represents an average difficulty for a particular item relative to the category intersections.

Finally, for each item and person the RUMM program computes an item-fit statistic. In the first set of columns in Table 5.6 are shown the item-fit values, their 1 degree of freedom chi-square values and the associated probabilities for the RSM. These values indicate that around half (5) of the 12 items do not fit this model ($p < .05$). We might have suspected that the RSM would yield a rather poor fit at the item level because our previous analysis with the G-PCM and GRM showed us that the items differ in slope. Simply stated, the NEO-FFI items probably do not form a rating scale as assumed by the RSM.

SCORE COMPARISONS

At this point, we thought it interesting to step back and examine what the examinee's latent trait scores look like under the various models described thus far. This process was facilitated because each of the programs used (i.e., RUMM and PARSCALE) provides for examinee scoring. All latent-trait scores reported here are maximum likelihood estimates with the

exception of the M-GRM in which we used EAP scoring and the GRM in which we used MAP scoring. These latter two scoring methods differ from ML primarily in their use of priors (see chap. 7 for details). There are six scores to be compared in these analyses: the raw scale score (RAW), and the latent trait estimates under the GRM, PCM, G-PCM, RSM, and M-GRM.

In Table 5.8 are shown the descriptive statistics for each of the scores estimated in the present study. These values differ across IRT models indicating the differing metrics that are used. Importantly, all latent trait scores are highly correlated (i.e., $r > .97$), and in turn, each is highly correlated with raw scores. In fact, the lowest correlation was $r = .97$ between raw scores and the G-PCM. These results are visually summarized by means of a scatterplot matrix (SPLOM) in Fig. 5.12. Of particular note here is that in the PCM and RSM the trait level estimates are a non-linear transformation of the raw scores (i.e., raw score is a sufficient statistic for trait level). This property does not hold true in any of the other models that contain a slope parameter.

The pattern of results shown in Fig. 5.12 should not be taken to mean that IRT modeling is therefore no better than simply computing a raw scale score. For a relative ordering of examinees, this may be true. However, the IRT models are optimal scalings of examinee trait levels and as described throughout this book, the IRT model parameters enable numerous psychometric advantages. For example, linking (equating) test scales, exploring differential item functioning, computing person-fit statistics, computerized adaptive testing, and allowing the standard error to change as a function of trait level are just some of psychometric advantages to the IRT modeling.

THE NOMINAL RESPONSE MODEL

Bock (1972) proposed a general divide-by-total or direct IRT model that can be used to characterize item responses when responses are not necessarily ordered along the trait continuum. In fact, all the divide-by-total or direct models described above are special cases of the nominal response model (Thissen & Steinberg, 1986). The original motivation for developing

TABLE 5.8
Descriptive Statistics for Raw Scores and Latent Trait Scores

	RAW	GRM	PCM	G-PCM	RSM	M-GRM
Mean	25.13	0.00	0.00	-0.01	0.11	0.00
SD	8.30	0.95	0.80	1.13	0.76	0.93
Skew	-0.13	-0.11	-0.16	-0.28	-0.15	-0.47
Kurtosis	-0.59	-0.07	0.31	0.88	0.27	-0.03

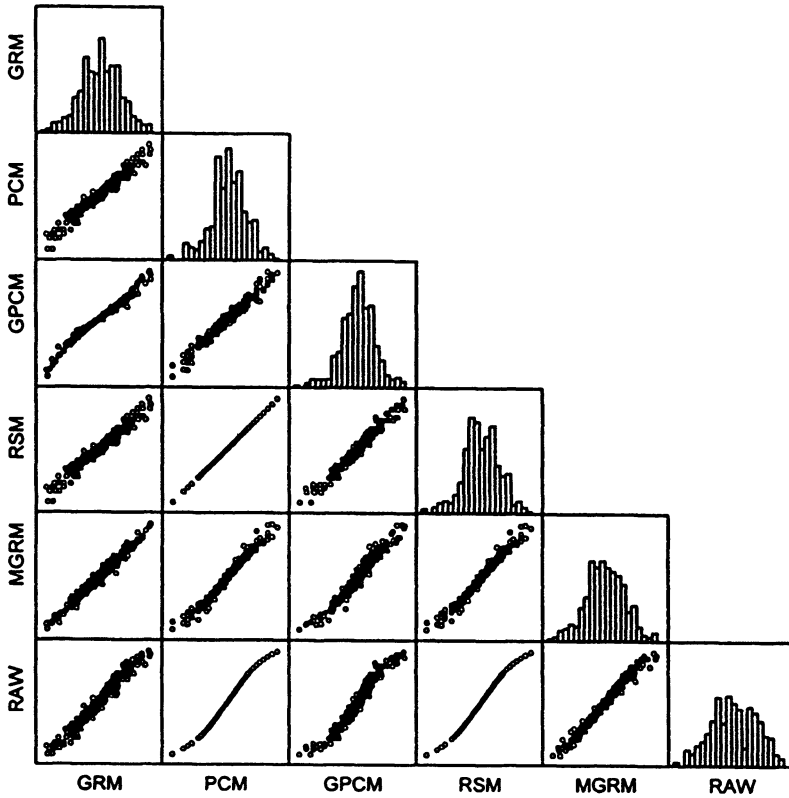


FIG. 5.12. SPLOM of trait level estimates under the various polytomous models.

the NRM was to allow the distracters in multiple-choice tests to be characterized by trace lines. Nevertheless, the NRM may be used for any items with multiple response options, which includes applications in personality and attitude assessment (see Thissen, 1993).

In the NRM, the probability of an examinee responding in category x ($x = 0 \dots m_i$) can be written as

$$P_{ix}(\theta) = \frac{\exp(\alpha_{ix}\theta + c_{ix})}{\sum_{x=0}^m \exp(\alpha_{ix}\theta + c_{ix})} \quad (5.11)$$

To identify the model (i.e., to estimate parameters), the constraint must be set that $\sum \alpha_{ix} = \sum c_{ix} = 0$, or in some cases, that the parameters for the lowest response category $\alpha_{i1} = c_{i1} = 0$. In this model, one α_{ix} and c_{ix} parameter

must be estimated for each of the $m_i + 1$ response categories within an item. The α_{ix} are related to the slope of the trace lines (i.e., the discrimination) for category x , and the c_{ix} is an intercept parameter for category x .

To illustrate, we borrow from an example given in Thissen (1993, p. 91). In this example, the items measure the presence or risk for bulimia. Most items within the scale contained sensible multiple ordered response categories. However, there was an item that had no obvious ordering of categories. Specifically, the item asked examinees to respond to the stem, "I prefer to eat:" with option choices of: (a) at home alone, (b) at home with others, (c) in a public restaurant, (d) at a friend's house, and (e) doesn't matter.

Thissen estimated the parameters of the nominal response model for this item and found $\alpha_a = -0.39$, $\alpha_b = -0.39$, $\alpha_c = 0.24$, $\alpha_d = -0.39$, $\alpha_e = 0.93$, and $c_a = 2.02$, $c_b = -1.49$, $c_c = -0.26$, $c_d = 0.57$, and $c_e = -0.83$. Figure 5.13 displays the response probabilities for this item. As evident in this figure, it is only response option (a), "at home alone," that is associated with increased risk for bulimia. Other examples of application of the nominal response model to psychological data may be found in Bock (1972) and in Thissen and Steinberg (1988).

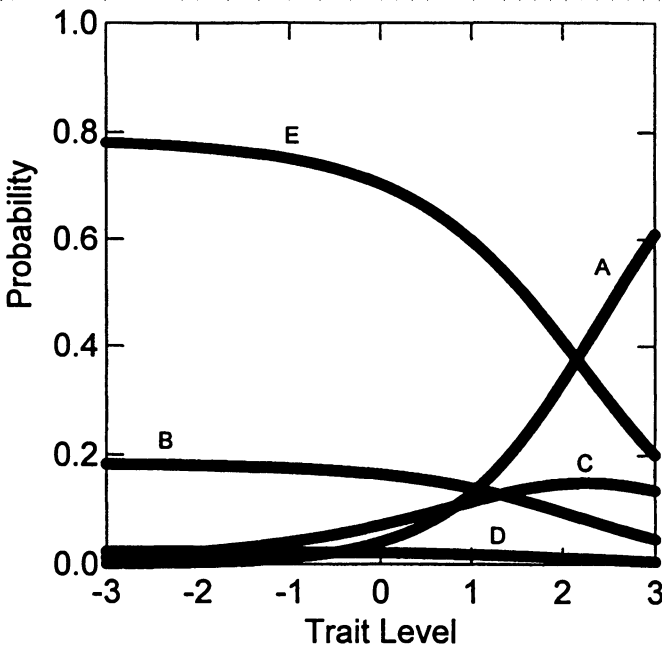


FIG. 5.13. Category response curves for the example under the nominal response model.

CONTINUOUS RESPONSE MODELS

Very rarely, researchers in the social sciences have made use of continuous response formats. For example, a researcher interested in attitudes may ask a subject to describe their views on a 1 to 100 scale, or a subject may be asked to indicate their views by marking on a line that is anchored from high to low. Furthermore, even the responses to Likert-type item formats can be viewed as approximations to a continuous scale. There is good reason for staying away from continuous self-report rating scales and multipoint Likert ratings in personality and attitude assessment; namely, they beg response bias due to the increased subjectivity of the expanded scale.

Nevertheless, as Mellenbergh (1994b) has pointed out, continuous response can be readily handled by IRT models by treating them as observed manifestations of a latent variable. In this case, factor analytic models may be used to understand item responses (see Thissen, Steinberg, Pyszczynski, & Greenberg, 1983). Working under this general framework, we note that Mellenbergh (1994b) has derived a latent-trait model for understanding continuous item responses and has demonstrated an application of the model in understanding a continuously rated adjective measure of neuroticism. Although we will not detail this model, we wish to alert readers to its existence and comment that it provides a framework for understanding IRT models and factor models as being among a variety of models falling under the umbrella of generalized linear item response models (Mellenbergh, 1994a).

MODELS NOT DESCRIBED

Due to space considerations, several potentially important models are not described in this chapter. However, we wish to mention these models so that interested readers can go to the original citations. For example, the multiple choice model (Thissen & Steinberg, 1984) may be used to characterize multiple choice test data, the successive intervals model (Rost, 1988) combines features of both the PCM and RSM, and Wilson's (1992) ordered-partitioned model, like the NRM, can handle test items where the response categories are not completely ordered. Finally, multidimensional extensions of polytomous IRT models have not been described, but see Kelderman and Rijkes (1994) for a log-linear multidimensional IRT parameterization, or Muraki and Carlson (1995) for a full-information item-factor analysis approach. Furthermore, it should be noted that several of the multidimensional models described in chapter 4 have polytomous generalizations (see original citations).

HOW MANY SUBJECTS?

Students and colleagues frequently ask how many subjects are needed to estimate IRT model parameters? There is no pat answer to this question because it depends on many factors, such as how discriminating the items are and how many item parameters are being estimated. In an attempt to resolve the number-of-subjects issue, some researchers have conducted Monte Carlo simulation studies. For example, Reise and Yu (1990) have shown that the GRM can be estimated with MULTILOG with as few as 250 examinees, but they recommend around 500. Yet simulation studies are useful only if the data matches the simulated data. An alternative approach is to recommend that a researcher have enough subjects to make the standard errors of parameter estimates reasonably small. The definition of "reasonably small," of course, depends on the researcher's measurement goals; the impact of item parameter estimates on trait level estimates is the ultimate issue for many researchers. As shown by the results here, some of the category threshold parameters were not well estimated in the GRM with 350 examinees.

There are some special considerations in estimating polytomous IRT models that researchers should note, and these are highly relevant to the number of subjects issue. First, it is highly recommended that a researcher estimate parameters based on a heterogeneous sample. Doing so should circumvent a second problem in fitting polytomous models, namely, a researcher must have item responses in each response category. If a response category has zero or few responses, a program to estimate parameters will not be able to estimate a good between category threshold or intersection. We have seen many samples where respondents do not choose a particular item category, possibly due to the quasi categorical nature of the trait. Of course, one way to circumvent this problem is to collapse across categories, or better, to rewrite the item to have fewer categories. However, see Andrich (1995) for some special considerations when collapsing categories.

MODEL COMPARISONS

Given the number of existing models for potential application, it is important to ask how a researcher should go about selecting a model to fit a real data set. Of course, one primary consideration is the type of data that a researcher is working with. Obviously, if response categories are not ordered, then the NRM is the only choice that is appropriate. If the items are suspected to have differing discriminations, as might be revealed through a preliminary factor analysis, then the GRM or one of the Muraki (1990;

1992) models is automatically more appropriate than any of the Rasch-based models such as the RSM or PCM. Also, if the measure does not use the same rating scale for each item, then the RSM is not applicable (unless blocks of items are analyzed separately).

As the reader will see in chapter 9, the science of statistical model comparison in IRT is deficit, especially when compared to structural equation models. Some researchers have used the log-likelihood of model-data fit as a vehicle for comparing different IRT models (e.g., Rost, 1988). Levine et al. (1992) presented an "ideal-observer" statistical technique that offers much promise for deciding if two models really differ in fit to real data. In applications of the ideal-observer technique to real polytomous data, Maydeu-Olivares, Drasgow, and Mead (1994) found very little difference in fit between the GRM and the Thissen and Steinberg (1986) extension of the PCM model.

If there is no readily available statistical magic wand to guide our choice of model selection, we are left to decide amongst models on the basis of more practical concerns. The first is the number of parameter estimates, models with fewer parameters are better than more parameters because, generally speaking, sample size needed for estimation will be smaller. Also, ease of scoring may be a concern. In Rasch models, like the PCM, the raw scale score is a sufficient statistic for examinee trait level, assuming all examinees have received the same items. Finally, one's philosophy may dictate model use. For some researchers, it is not appropriate to demand that test items have equal discrimination, and they will use the GRM as opposed to one of the Rasch models. Others researchers may prefer to have scale difficulties such as in the RSM model. In fact, attitude researchers may find the RSM very attractive. Also, it goes without saying that models will only be fit if appropriate and usable programs are readily available.